



IoT Greengrass – Overview

What is AWS IoT Greengrass?

AWS IoT Greengrass extends AWS capabilities to **edge devices**, enabling **local data processing, messaging, machine learning (ML), and device control** even when not connected to the cloud.

 Greengrass = "AWS Lambda + MQTT + Local Resource Access" **at the edge**

Key Components of AWS IoT Greengrass

Component	Description
 Greengrass Core	Installed on edge devices (Raspberry Pi, industrial gateways, etc.) to run local apps
 IoT Thing	A registered IoT device in AWS IoT Core
 Components	Modular building blocks like Lambda functions, containers, or custom apps
 Local Shadow Service	Local replica of IoT device state (syncs with AWS IoT Core)
 Local MQTT Broker	Facilitates offline device-to-device messaging
 Greengrass CLI & Deployment Tool	For managing components and configurations

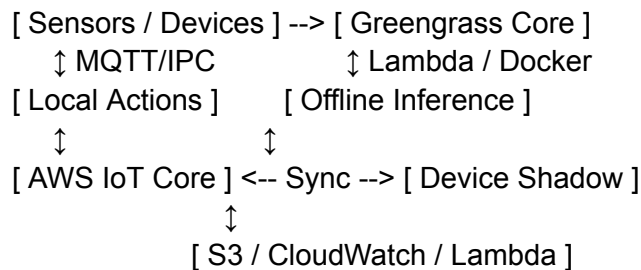
What Can It Do?

- ✓ Run **Lambda functions** at the edge
 - ✓ Execute **Docker containers** or binaries locally
 - ✓ Use **local messaging** via MQTT
 - ✓ Sync with **cloud device shadow**
 - ✓ Perform **machine learning inference** offline
 - ✓ Connect to **local resources** (camera, serial devices, sensors)
 - ✓ Collect and forward telemetry or logs to the cloud
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How It Works

1. You deploy **Greengrass Core software** on a device (e.g., industrial computer)
 2. It securely connects to **AWS IoT Core**
 3. You define and deploy **components** (apps, ML models, scripts)
 4. Device runs logic **locally**, and syncs to cloud when online
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Sample Architecture



Component Examples

Component Type	Example
 Lambda	Read sensor and log locally
 ML Inference	Run model for defect detection

 Docker	Legacy industrial control system
 Local Resource Access	GPIO, USB camera, serial port
 MQTT Pub/Sub	Device-to-device communication
 Cron Jobs	Run health check scripts periodically

Security Features

Feature	Description
TLS Mutual Auth	Between Greengrass and AWS IoT Core
Device Identity	X.509 certificate-based
Role Alias	IAM role assumption from edge
Component Isolation	Runtime isolation, resource permissions

Real-World Use Cases

Industry	Use Case
 Manufacturing	Machine control, quality detection
 Agriculture	Local crop/soil data processing
 Automotive	In-vehicle data processing, OTA updates
 Smart Cities	Traffic light control, surveillance
 Healthcare	Local vitals monitoring, alerting
 Retail	In-store checkout systems, edge ML

Pricing Overview

Service	Pricing
AWS IoT Greengrass	Free to install and run on devices
Components (e.g., Lambda, S3, etc.)	Billed as per usage in the cloud
OTA Updates, Cloud Storage, Data Transfer	Based on respective AWS service pricing

✔ No charge for Greengrass software itself; pay for cloud services used.

✔ Benefits

- ✔ Run workloads **locally at the edge**
 - ✔ **Offline operation** possible
 - ✔ Tight **integration with AWS** services
 - ✔ Manage fleets of devices centrally
 - ✔ Supports **ML, Docker, custom code**
 - ✔ **Secure by design** (certs, policies)
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! Challenges / Considerations

- Learning curve for **component modeling**
 - Network and cert setup complexity
 - Some features require **device resources (RAM/CPU)**
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🔧 Would You Like a Lab?

I can build a **Greengrass demo lab** for you:

🔧 Lab Idea: Local Temperature Monitor

- Deploy Greengrass Core on a Raspberry Pi

- Use Lambda to read temp sensor every 10s
- Publish to local MQTT topic
- Sync to AWS IoT Core when back online
- Trigger alert via SNS if temperature exceeds threshold

Summary Table

Feature	Description
 Local compute	Run code, ML, Docker at the edge
 Local messaging	MQTT + IPC
 Offline sync	Device Shadows
 ML Inference	Deploy and run pre-trained models
 Security	TLS, IAM, X.509, isolation
 Pricing	Pay-as-you-go (only for AWS services used)
